

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
 ORGANISATION OF ISLAMIC COOPERATION (OIC)
 DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Semester Final Examination
 Course No.: PHY 4241
 Course Title: Physics II

Summer Semester: 2021 - 2022
 Full Marks: 150
 Time: 3 hrs.

There are 06 (six) questions. Answer all 06 (six) questions. The symbols have their usual meanings. Marks of each question and corresponding CO and PO are written in brackets.

1. i. Find I_s in the circuit in Fig. 1(a). Here, $V_0 = 8\angle 30^\circ$.

8

(CO2)

(PO1, PO2)

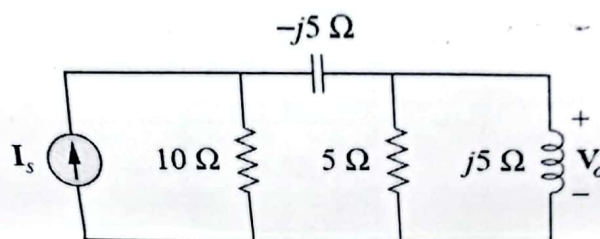


Fig. 1(a)

- ii. Find V_s in the circuit in Fig. 1(b). Here, $I_0 = 2\angle 0^\circ$.

8

(CO2)

(PO1, PO2)

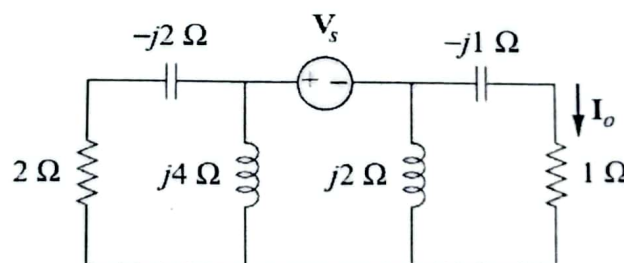


Fig. 1(b)

- iii. For the circuit in Fig. 1(c), find Z_{eq} .

9

(CO2)

(PO1, PO2)

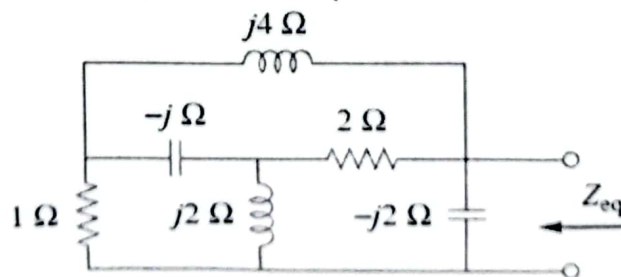


Fig. 1(c)

2. i. Find V_0 for the circuit in Fig. 2(a) using Nodal analysis.

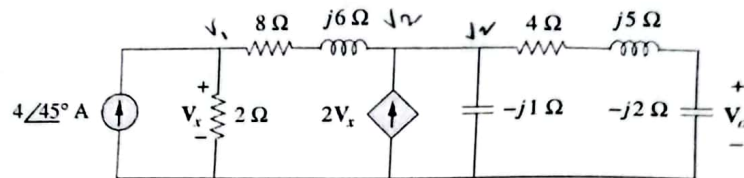


Fig. 2(a)

- ii. Find I_0 for the circuit in Fig. 2(b) using Mesh analysis.

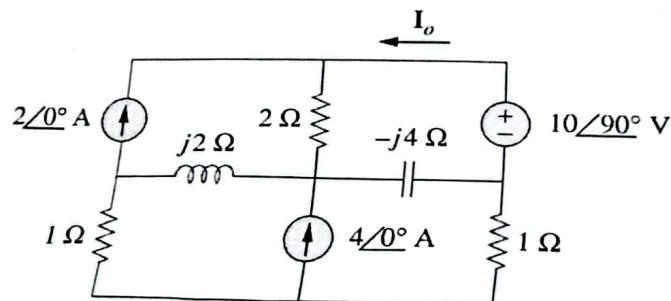


Fig. 2(b)

12
(CO2)
(PO1, PO2)

13
(CO2)
(PO1, PO2)

3. i. Find i_0 for the circuit in Fig. 3(a).

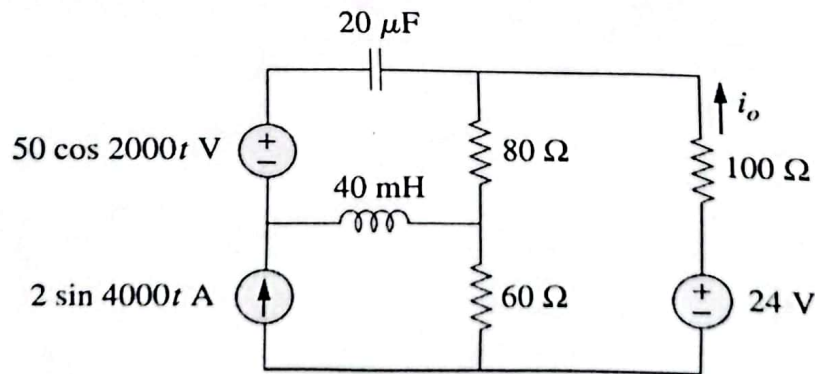


Fig. 3(a)

12
(CO2)
(PO1, PO2)

- ii. Find the Thevenin equivalent at terminal a-b in the circuit of Fig. 3(b).

13
(CO2)
(PO1, PO2)

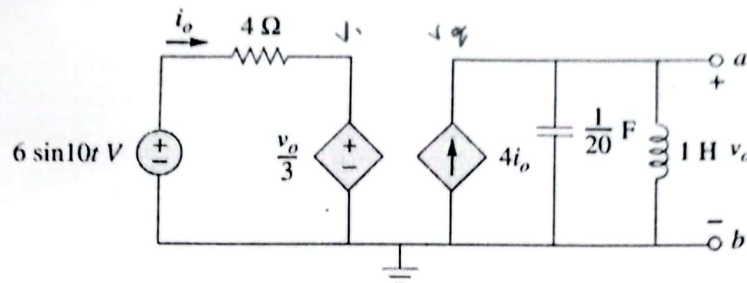


Fig. 3(b)

4. i. Find C_{eq} for the circuit in Fig. 4(a). All capacitors are $4 \mu\text{F}$.

8
(CO2)
(PO1, PO2)

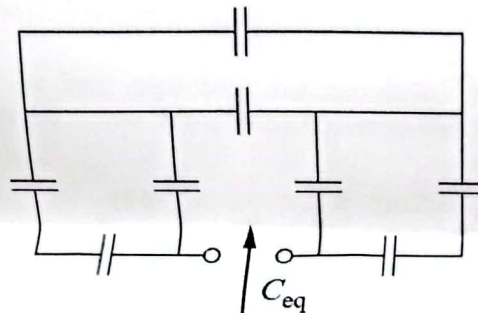


Fig. 4(a)

- ii. Find voltages across the capacitors for the circuit in Fig. 4(b).

8
(CO2)
(PO1, PO2)

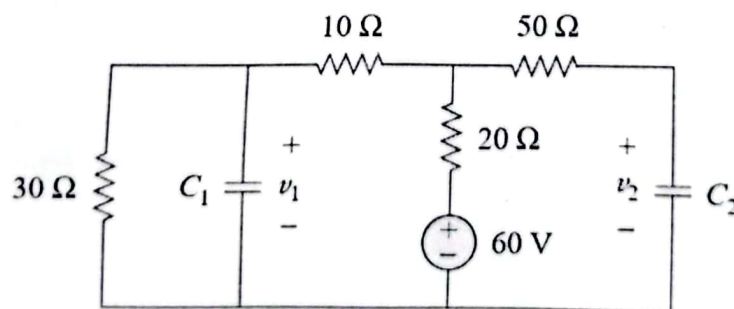


Fig. 4(b)

$$\frac{j\omega}{20000} + \frac{1}{j\omega} = \frac{j\omega 20000 + 20000}{4000000}$$

- iii. For the circuit in Fig. 4(c), find L_{eq} , $i_1(t)$, $i_2(t)$, $v_o(t)$, and energy stored in the 20-mH inductor at 1s. Here, $i_s(t) = 3e^{-t}mA$.

9
(CO2)
(PO1, PO2)

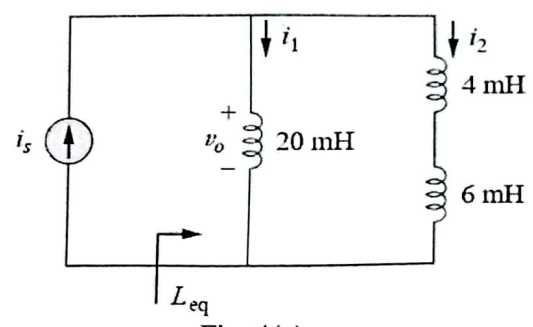


Fig. 4(c)

$$i(0) = 0$$

5. i. Determine the resonant frequency for the circuit in Fig. 5(a).

8
(CO2)
(PO1, PO2)

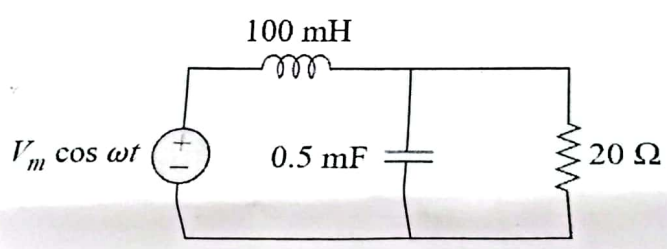


Fig. 5(a)

- ii. Determine ω_0 , Q , and B for the circuit in Fig. 5(b). Here, $v_o = 20 \cos \omega t$.

8
(CO2)
(PO1, PO2)

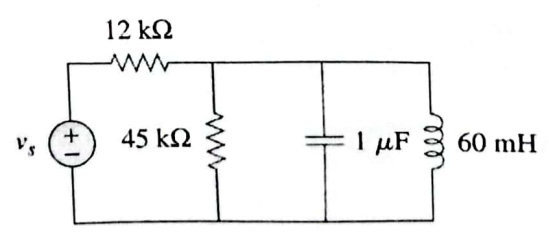


Fig. 5(b)

- iii. Find the transfer function for the circuit in Fig. 5(c) and determine what type of filter it represents.

9
(CO2)
(PO1, PO2)

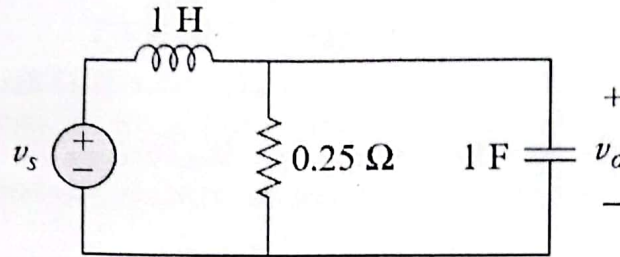


Fig. 5(c)

6. i. Design a BPF using a series RLC network that has a lower cut-off frequency of 20.1 KHz and an upper cut-off frequency of 20.3 KHz. Take $R=20\text{K}\Omega$. Determine L, C and Q.
- ii. Determine the load impedance that maximizes the average power drawn from the circuit in Fig. 6(b). Calculate the maximum average power.

8
(CO2)
(PO1, PO2)

8
(CO2)
(PO1, PO2)

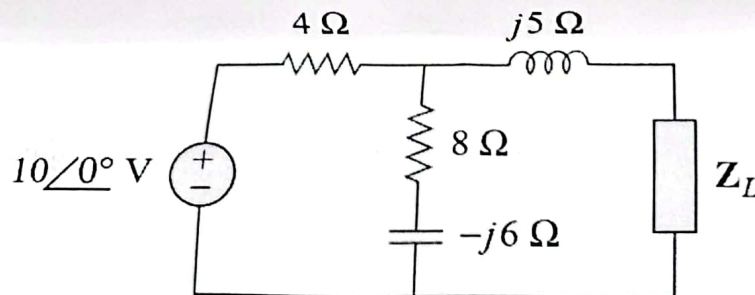


Fig. 6(b)

- iii. Explain the concept of power flow in ac circuit and the significance of power factor. Draw the power triangle and explain complex power, real power and reactive power.

9
(CO2)
(PO1, PO2)